

Case Study

Assessment week 1 | case study 2

Retail Sales & Customer Insights Dashboard-Cloud That

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(Batch 3)



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- **Links**

- GitHub : [Case_Study/W1_Case_Study_2 at main · amankumar1ext-web/Case_Study](https://github.com/amankumar1ext-web/Case_Study)

CHAPTER 1: INTRODUCTION

1.1 Introduction

In today's competitive retail industry, organizations generate large volumes of transactional and customer-related data from multiple sources. However, raw data alone does not provide meaningful insights for business decision-making. Retail companies often face challenges in tracking sales performance, identifying profitable products, understanding customer purchasing behavior, and analyzing regional market trends.

To address these challenges, data warehousing and business intelligence solutions have become essential. A centralized data warehouse allows organizations to integrate data from multiple systems, transform it into a structured format, and generate meaningful insights using analytical tools.

The Retail Sales & Customer Insights Dashboard project focuses on collecting sales, customer, and product information from multiple data sources and transforming it into a unified analytical platform. The project combines Data Warehousing, ETL processes, SQL analytics, Python-based data preprocessing, and Power BI visualization to provide valuable business insights.

The final dashboard enables stakeholders to monitor sales trends, identify top-performing products, analyze customer demographics, evaluate regional performance, and make data-driven business decisions.

1.2 Background of the Project

Retail organizations operate across multiple stores and regions, generating data from different operational systems. These systems often include:

- Transactional sales databases
- Customer management systems
- Product information repositories
- External business datasets

Because data is stored in different formats and locations, decision-makers face difficulties obtaining a consolidated view of business performance.

The Retail Sales & Customer Insights Dashboard was developed to solve this issue by integrating:

- Sales Data from MySQL
- Customer Data from NoSQL Database
- Product Data from SharePoint/CSV Source

The integrated data is transformed through an ETL pipeline and stored in a data warehouse designed using a Star Schema model. This centralized structure improves reporting efficiency and analytical capabilities.

1.3 Problem Statement

A retail chain operating across multiple locations faces challenges in effectively analyzing its growing volume of sales and customer data. The organization struggles to identify top-performing products, understand customer purchasing behavior, evaluate regional sales performance, and make timely data-driven decisions.

The absence of a centralized analytics platform results in:

- Difficulty identifying best-selling products.
- Lack of visibility into customer purchasing patterns.
- Inability to understand seasonal trends and promotional impacts.
- Challenges in regional sales comparison.
- Limited support for inventory optimization.
- Inefficient marketing decision-making processes.

Therefore, there is a need to design and implement a data-driven business intelligence solution capable of integrating multiple data sources, storing information in a centralized data warehouse, and presenting actionable insights through interactive visual dashboards.

1.4 Project Objectives

The primary objective of this project is to develop a Retail Sales & Customer Insights Dashboard that enables effective analysis of sales and customer data.

Specific Objectives

1. To integrate data from multiple heterogeneous sources such as MySQL, NoSQL databases, and SharePoint files.
2. To design and implement a Data Warehouse using a Star Schema architecture.
3. To develop an ETL pipeline for extracting, transforming, and loading data into the warehouse.
4. To perform data cleaning and quality validation using Python.
5. To analyze data using SQL queries and generate meaningful business insights.
6. To calculate key business KPIs such as:
 - Total Sales Revenue
 - Sales Growth Rate
 - Customer Lifetime Value (CLV)
 - Regional Sales Performance
 - Top Selling Products
7. To build an interactive Power BI dashboard that supports business decision-making.
8. To provide visualizations and reports that help stakeholders understand customer behavior and sales performance.

1.5 Scope of the Project

The scope of this project includes:

Data Collection

Collection of data from multiple business systems including sales, customer, and product datasets.

Data Warehousing

Design and implementation of a centralized data warehouse structure capable of storing integrated business data.

ETL Process

Developing data extraction, transformation, and loading mechanisms to ensure consistency and reliability.

Data Analysis

Performing SQL-based analysis to identify trends, customer segments, and sales patterns.

Dashboard Development

Creating interactive Power BI dashboards that provide:

- Sales overview
- Product performance analysis
- Customer insights
- Regional analysis
- KPI monitoring

Business Intelligence

Generating actionable insights for management to support strategic and operational decision-making.

1.6 Expected Outcomes

The successful implementation of this project is expected to provide:

- Improved visibility into business performance.
- Better understanding of customer behavior.
- Identification of top-selling products.
- Enhanced regional sales analysis.
- Faster access to business insights through automated reporting.
- Improved inventory and marketing decision-making.
- A scalable analytics solution that can support future business growth.

1.7 Technologies Used

TECHNOLOGY	PURPOSE
MYSQL	Sales Data Storage
NOSQL DATABASE	Customer Data Storage
SHAREPOINT / CSV	Product Data Source
PYTHON	Data Cleaning and Transformation
SQL	Data Analysis and Query Development
DATA WAREHOUSE	Centralized Data Storage
POWER BI	Dashboard Development and Visualization

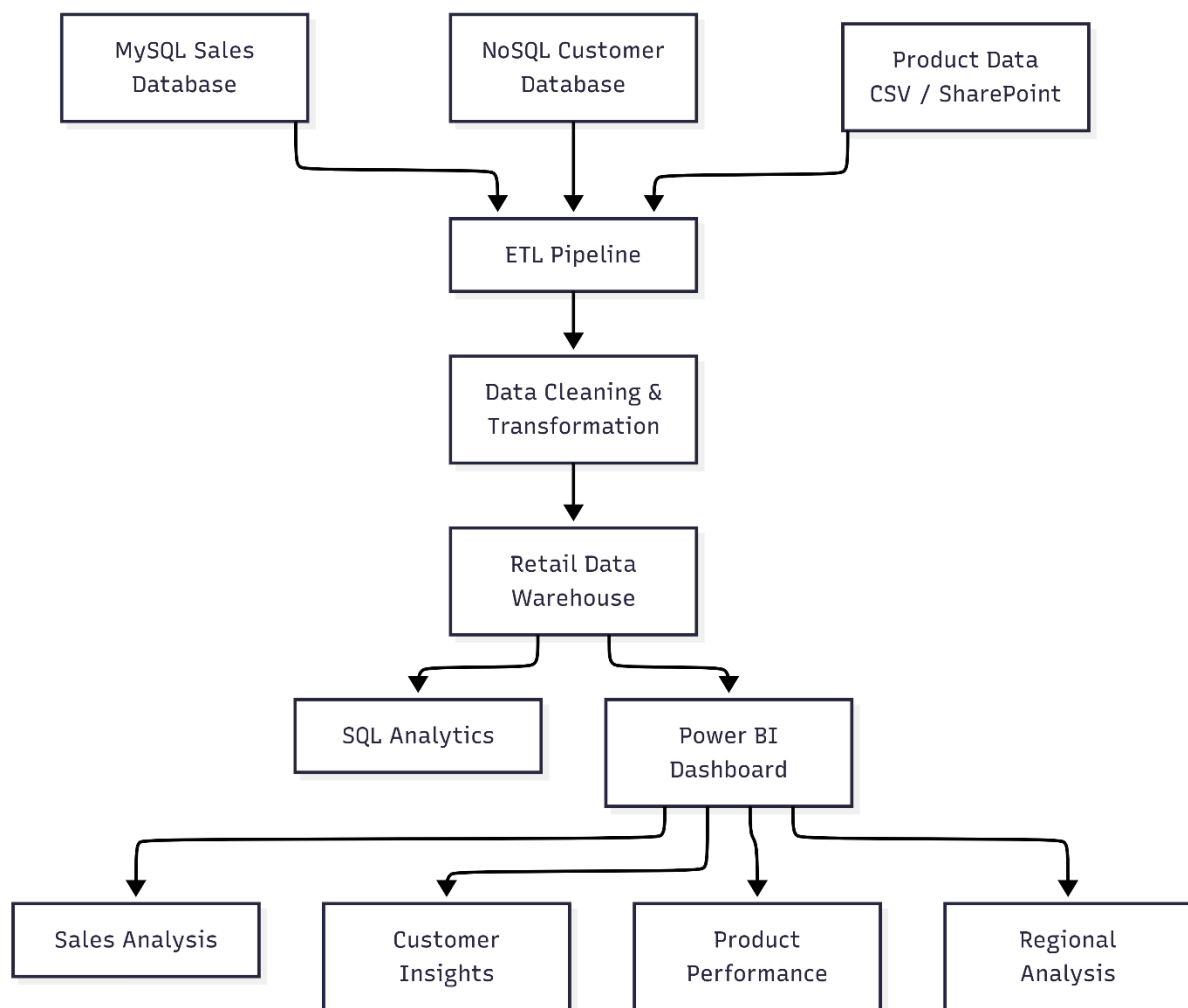


Figure 0.1 Overall Architecture of the Retail Sales & Customer Insights Dashboard System

The overall architecture illustrates the flow of data from multiple sources including MySQL sales databases, NoSQL customer repositories, and product datasets stored in CSV or SharePoint. The data is processed through an ETL pipeline where cleaning and transformation activities are performed. The transformed data is loaded into the Retail Data Warehouse, which serves as the centralized repository for analytics. SQL is used to perform analytical queries, while Power BI is used to create interactive dashboards and visual reports for business stakeholders.

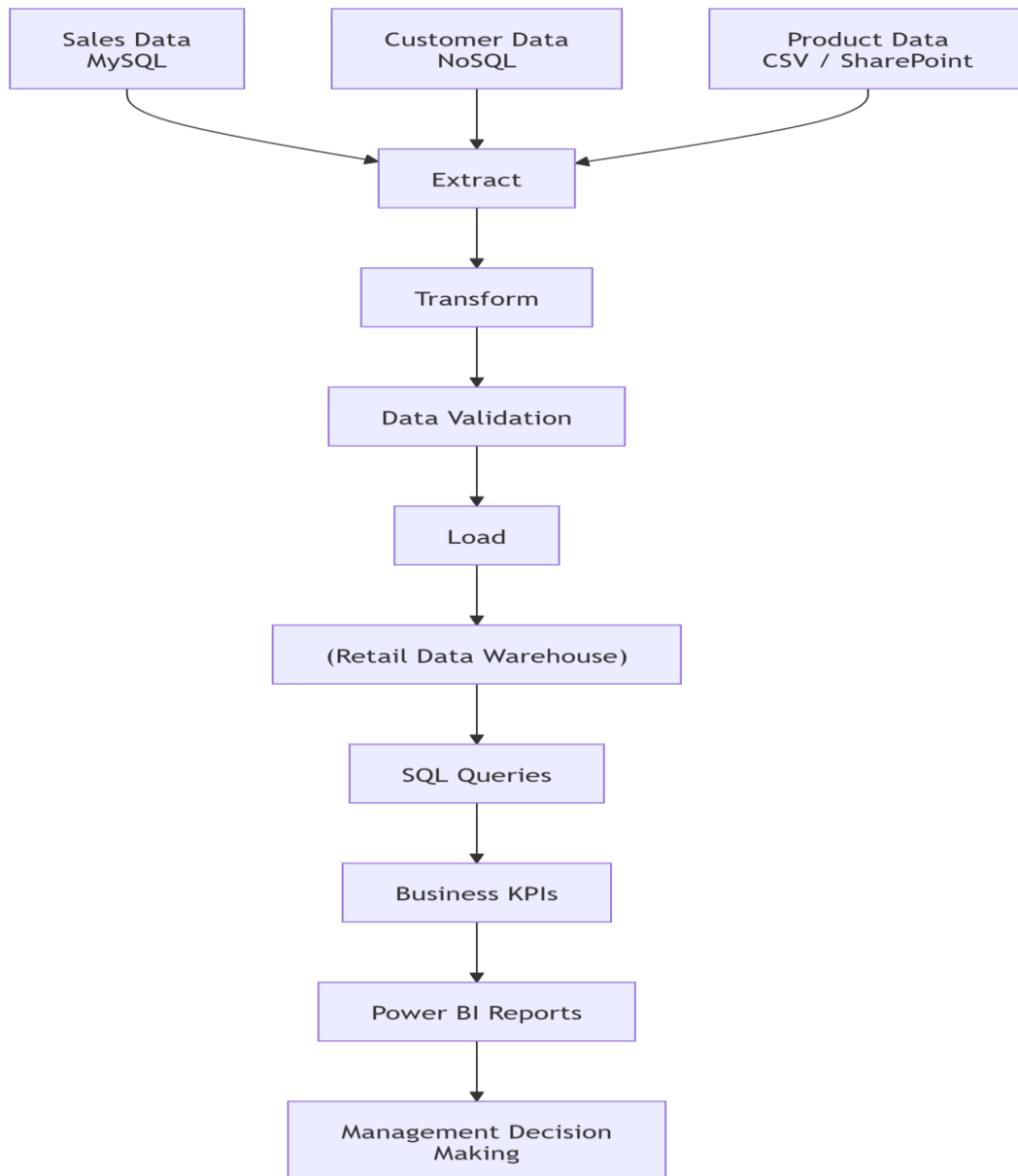


Figure 0.2: Data Flow Diagram Showing ETL Process and Dashboard Generation

The Data Flow Diagram represents the movement of data throughout the system. Data is extracted from three independent sources, transformed into a consistent format, validated for quality and accuracy, and loaded into the central data warehouse. Analytical SQL queries are executed on the warehouse to calculate business KPIs and generate insights. These insights are visualized using Power BI dashboards that support business decision-making.

1.8 Report Organization

This report is organized into the following chapters:

- Chapter 1: Introduction
- Chapter 2: Requirement Analysis and Data Understanding
- Chapter 3: Data Warehouse Design
- Chapter 4: ETL Development and Data Integration
- Chapter 5: Data Cleaning and SQL Analysis
- Chapter 6: Power BI Dashboard Development
- Chapter 7: Results and Insights

CHAPTER 2: DATA UNDERSTANDING & REQUIREMENT ANALYSIS

2.1 Dataset Overview

The project uses three datasets representing different business entities:

1. Sales Dataset ([sales 1](#))
2. Customer Dataset ([customers](#))
3. Product Dataset ([products](#))

These datasets were integrated to create a unified retail analytics solution.

2.2 Sales Dataset Analysis

The sales dataset contains transactional records including:

- SaleID
- ProductID
- CustomerID
- SalesAmount
- Quantity
- Timestamp

Sample records show missing SalesAmount and Quantity values in some transactions.

A	B	C	D	E	F	G
SaleID	ProductID	CustomerID	SalesAmount	Quantity	Timestamp	
c0078cd1-	1020	C0460	1639.1	18	2024-08-09T20:21:00	
be8532af-	1029	C0198	186.11	42	2024-11-28T05:01:00	
8666f277-	1012	C0524	1712.72	49	2024-10-13T00:36:00	
383e0a2b-	1016	C0285		50	2024-07-06T23:39:00	
d9a7f974-	1021	C0369	66.36	21	2024-06-08T19:58:00	
d71733ac-	1016	C0709	1705.51	31	2024-11-15T12:24:00	
b7475afb-	1047	C0929	1509.94	3	2024-08-28T05:43:00	

Figure 0.1: Sales Data Missing Value

Missing values and inconsistent records were identified, indicating the necessity of data cleaning.

2.3 Customer Dataset Analysis

The customer dataset contains:

- CustomerID
- FirstName
- LastName
- Gender
- Region
- SSN

The data contains missing values and inconsistent region names such as "Ohio", "Ohho", "NY", "New Yorkk", and "Nw York".

2.4 Product Dataset Analysis

The product dataset contains:

- ProductID
- ProductName
- Category

Products are grouped under:

- Electronics
- Furniture
- Home Appliances
- Toys

A	B	C	D
ProductID	ProductNa	Category	
1000	Washing M	Home Appliances	
1001	Refrigerato	Home Appliances	
1002	Blender	Home Appliances	
1003	Microwave	Home Appliances	
1004	Vacuum C	Home Appliances	
1005	Sofa	Furniture	
1006	Table	Furniture	
1007	Chair	Furniture	
1008	Bed	Furniture	
1009	Cupboard	Furniture	
1010	Laptop	Electronics	
1011	Smartphor	Electronics	
1012	Headphon	Electronics	

Figure 0.2:Product Data

Product data is highly structured and requires minimal cleansing before integration.

2.5 Business Requirements

The dashboard should answer:

- Which products generate highest sales?
- Which regions perform best?
- What are monthly sales trends?
- Which customers contribute most revenue?
- How does category performance vary?

2.6 Data Quality Issues Identified

ISSUE	DATASET
MISSING SALESAMOUNT	Sales
MISSING QUANTITY	Sales
MISSING CUSTOMER DETAILS	Customers
INCONSISTENT REGION NAMES	Customers
INCONSISTENT GENDER VALUES	Customers
OUTLIER SALES VALUES	Sales

Chapter 3: Data Warehouse Design

3.1 Schema Design

A Star Schema model was designed to organize the retail sales data for analytical reporting. The model consists of one fact table and multiple dimension tables.

3.2 Star Schema Diagram

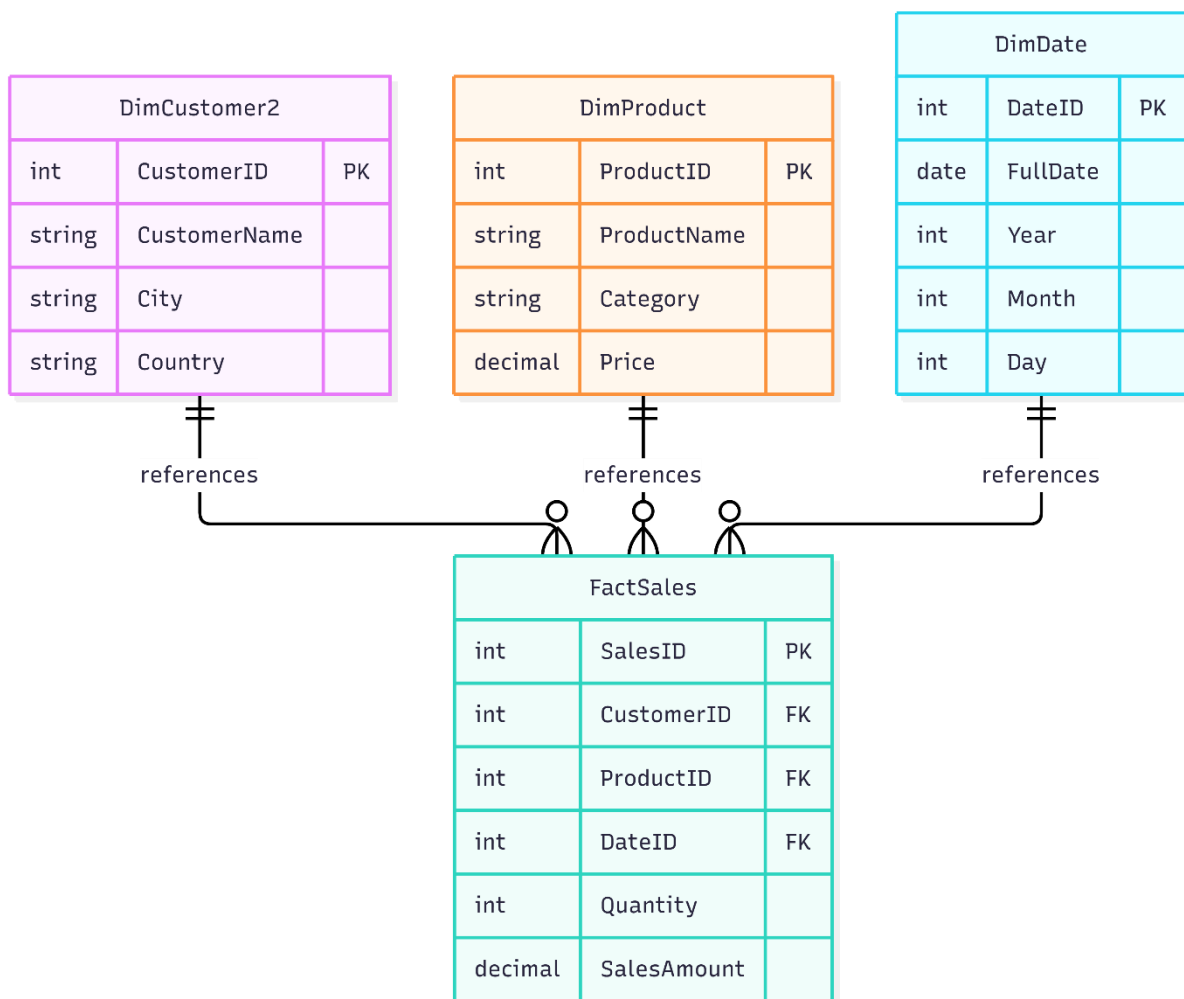


Figure 0.1: Star Schema Diagram

3.3 Conclusion

The retail data warehouse was designed using a Star Schema consisting of FactSales, DimCustomer, DimProduct, and DimDate tables. This structure provides a simple and efficient model for SQL analysis and Power BI dashboard development.

Chapter 4: ETL Development and Data Integration

4.1 Data Extraction

Data was extracted from three different sources:

DATASET	SOURCE
SALES DATA	CSV / MySQL Export
CUSTOMER DATA	JSON (NoSQL)
PRODUCT DATA	CSV

The datasets were imported into Python using the Pandas library for data preprocessing and quality validation. The objective was to identify missing values, inconsistencies, duplicate records, and formatting issues before loading data into the data warehouse.

```
print(customers.head())
```

	CustomerID	FirstName	LastName	Gender	Region	SSN
0	C0001	Gregory	Miller	M	Ohho	082573197
1	C0002	Marvin	NaN	male	california	309-6256-21
2	C0003	Gregory	Smith	Male	New Yorkk	407071725
3	C0004	Edward	Davis	male	Ohho	948.18.5740
4	C0005	Reginald	Dawson	male	Texas	975.72.3056

```
print(products.head())
```

	ProductID	ProductName	Category
0	1000	Washing Machine	Home Appliances
1	1001	Refrigerator	Home Appliances
2	1002	Blender	Home Appliances
3	1003	Microwave	Home Appliances
4	1004	Vacuum Cleaner	Home Appliances

Figure 0.1: Data Imported In Python

4.2 Data Profiling

Data profiling was performed to understand the structure, data types, missing values, and overall quality of each dataset.

```
[12]: # Duplicate records
print("Sales Duplicates:", sales.duplicated().sum())
print("_____")
print("Customer Duplicates:", customers.duplicated().sum())
print("_____")
print("Product Duplicates:", products.duplicated().sum())
```

Sales Duplicates: 0

Customer Duplicates: 100

Product Duplicates: 0

Figure 0.2: Duplicates Records

4.3 Data Cleaning

Several data quality issues were identified during profiling. Missing numerical values, missing customer attributes, and inconsistent region names were corrected to improve data accuracy and consistency.

	CustomerID	FirstName	LastName	Gender	Region	SSN
3	C0004	Edward	Davis	male	Ohio	948.18.5740
4	C0005	Reginald	Dawson	male	Texas	975.72.3056

```
[26]: customers.nunique()

[26]: CustomerID    1000
      FirstName    318
      LastName     400
      Gender        4
      Region        6
      SSN          1000
      dtype: int64
```

Region Standardization

```
[28]: customers["Region"] = customers["Region"].replace({
      "Ohho": "Ohio",
      "OHIO": "Ohio",
      "NY": "New York",
      "New Yorkk": "New York",
      "Nw York": "New York"
    })
```

Figure 0.3: Region Standardization

```
[29]:
```

	CustomerID	FirstName	LastName	Gender	Region	SSN
0	C0001	Gregory	Miller	Male	Ohio	082573197
1	C0002	Marvin	NaN	male	california	309-6256-21
2	C0003	Gregory	Smith	Male	New York	407071725
3	C0004	Edward	Davis	male	Ohio	948.18.5740
4	C0005	Reginald	Dawson	male	Texas	975.72.3056

```
[30]: customers["Region"] = customers["Region"].fillna("Unknown")

[ ]:
```

```
[31]: sales = sales.drop_duplicates()
      customers = customers.drop_duplicates()
      products = products.drop_duplicates()

[32]: sales.count()

[32]: SaleID          10000
      ProductID     10000
      CustomerID    10000
      SalesAmount    10000
```

Figure 0.4: Remove Duplicate

4.4 Outlier Detection

Outlier analysis was performed on SalesAmount to identify unusually high or low transaction values that could impact business analysis.

```
Q1 = sales["SalesAmount"].quantile(0.25)
Q3 = sales["SalesAmount"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = sales[
    (sales["SalesAmount"] < lower) |
    (sales["SalesAmount"] > upper)
]

print(outliers.shape)

(79, 6)

print(outliers)
```

	SaleID	ProductID	CustomerID	SalesAmount	\
179	f88e997a-65bc-4e3c-966f-7d8796f83c5d	1029	C0256	40458.92	
483	f01892e4-149f-40a1-95e7-fda3df4d0bc5	1032	C0349	64357.03	
542	b2a09161-e694-47ca-bbc5-cc74e30b3121	1045	C0064	51858.11	
690	d2f8521f-9fde-435a-8a50-2ea238052ab2	1047	C0041	21176.88	
787	60337106-faa0-48a5-b55d-75db312c71ab	1004	C0426	61821.50	
...	...	...	...	...	
9601	2fcaa635-5d6a-4bd5-bdc1-876b86c444da	1030	C0144	61695.49	
9805	dec7ebdc-ea35-4dcb-9e97-688b640307b1	1034	C0122	59637.24	
9927	bae96876-7a12-489a-91d3-3d8792b91c3e	1030	C0269	76993.72	
9937	56aa51cf-b202-4a01-8f5e-5b5a335ceba7	1039	C0051	77834.11	
9957	45f3cc21-5244-4dda-8f0d-c993de9a53c4	1005	C0009	27261.56	

Figure 0.5: Outliers

4.5 Data Transformation

The cleaned data was transformed into a consistent format suitable for warehouse loading. Date fields were converted into datetime format and additional analytical attributes were created.

4.6 Export Cleaned Data

After successful cleaning and transformation, the processed datasets were exported and prepared for loading into the data warehouse.

Chapter 5: SQL Warehouse Tables Creation and Loading Process

5.1 Objective

The cleaned datasets generated through Python ETL were loaded into a relational database to create the data warehouse structure for reporting and analysis.

5.2 Database Creation

A separate Database will be created for this task.

5.3 Dimension Tables Creation

DimCustomer : The DimCustomer table stores customer demographic information used for customer segmentation and regional analysis.

DimProduct : The DimProduct table stores product details including category and pricing information.

DimDate : The DimDate table stores date-related attributes to support time-based analysis.

5.4 Fact Table Creation

The FactSales table stores transactional sales records and maintains relationships with dimension tables through foreign keys.

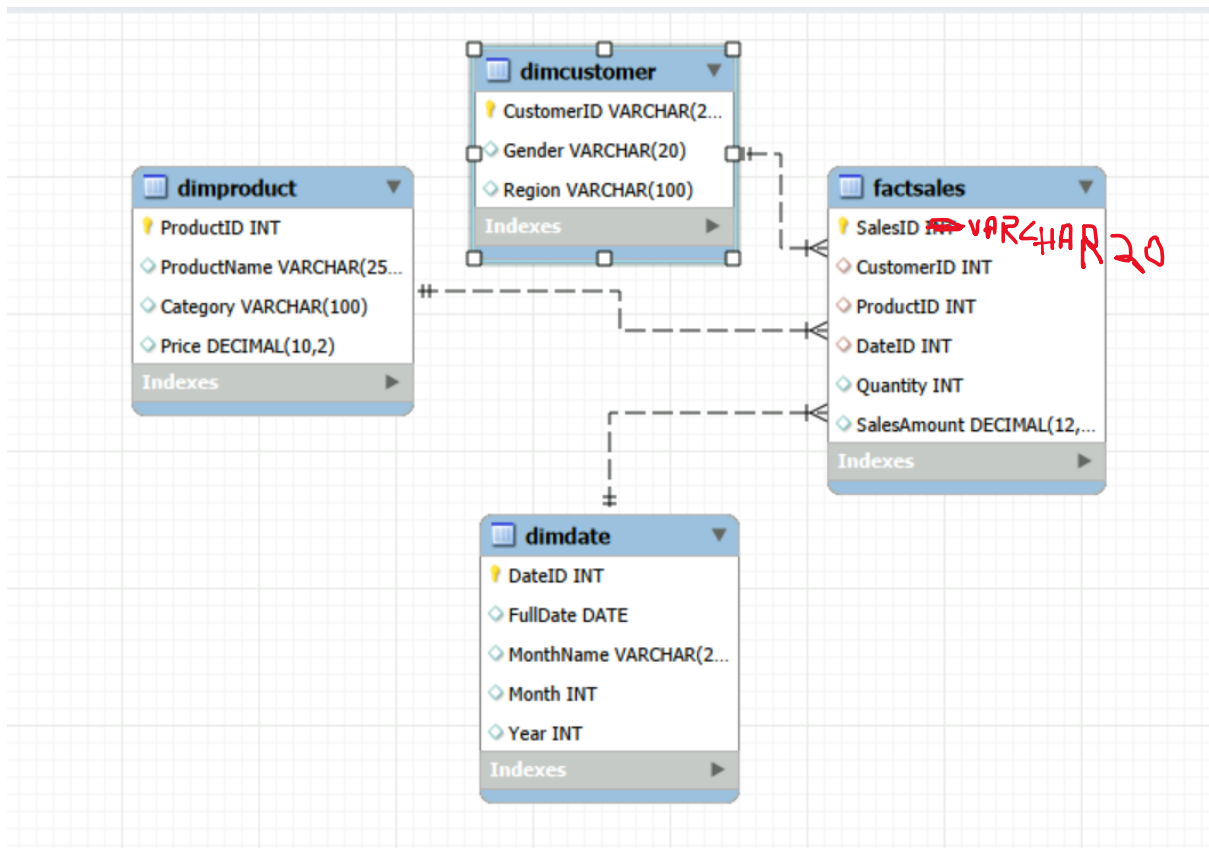


Figure 0.1: Data Model In SQL

5.5 Data Loading

The cleaned CSV files were imported into the warehouse tables.

5.7 Chapter Conclusion

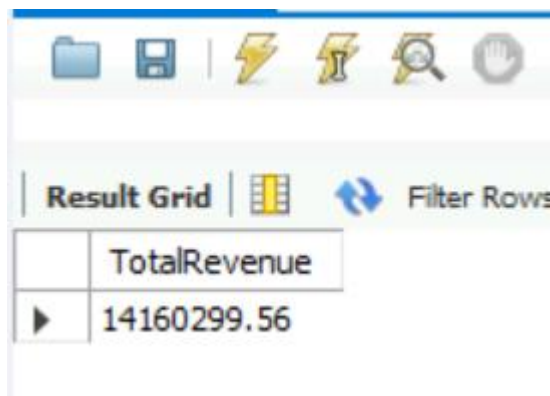
The cleaned datasets were successfully loaded into the RetailDW database.

Chapter 6: SQL Analysis and Business Queries

6.1 Overview

After loading the cleaned data into the warehouse, SQL queries were executed to analyze sales performance, customer behavior, and product trends. The results were subsequently used for Power BI dashboard development.

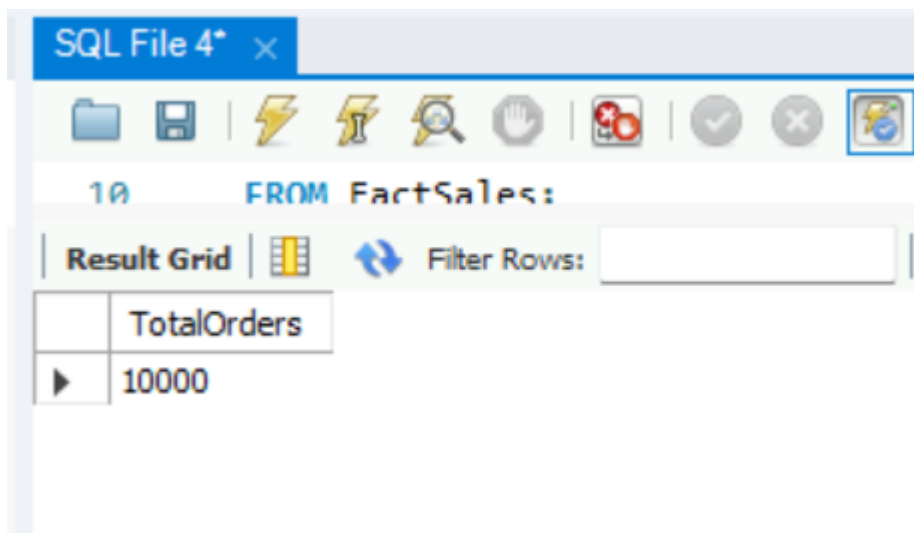
6.2 Total Revenue



The screenshot shows a SQL query result grid. The top toolbar includes icons for file operations, execution, and search. Below the toolbar, the 'Result Grid' tab is active, displaying a single row with the column 'TotalRevenue' and the value '14160299.56'. A 'Filter Rows' button is visible to the right of the grid.

	TotalRevenue
▶	14160299.56

6.3 Total Orders



The screenshot shows a SQL query result grid in a window titled 'SQL File 4*'. The toolbar includes icons for file operations, execution, search, and a 'Filter Rows' button. The query text '10 FROM FactSales:' is visible above the grid. The 'Result Grid' tab is active, displaying a single row with the column 'TotalOrders' and the value '10000'.

	TotalOrders
▶	10000

6.4 Revenue by Region

```
16 FROM FactSales f
17 JOIN DimCustomer c
18 ON f.CustomerID = c.CustomerID
19 GROUP BY c.Region
20 ORDER BY Revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Region	Revenue
▶	New York	5440075.07
	Ohio	2510945.53
	Texas	1377572.29
	Texaz	1373990.84
	Californiya	1353841.45
	california	1154170.63
	Unknown	949703.75

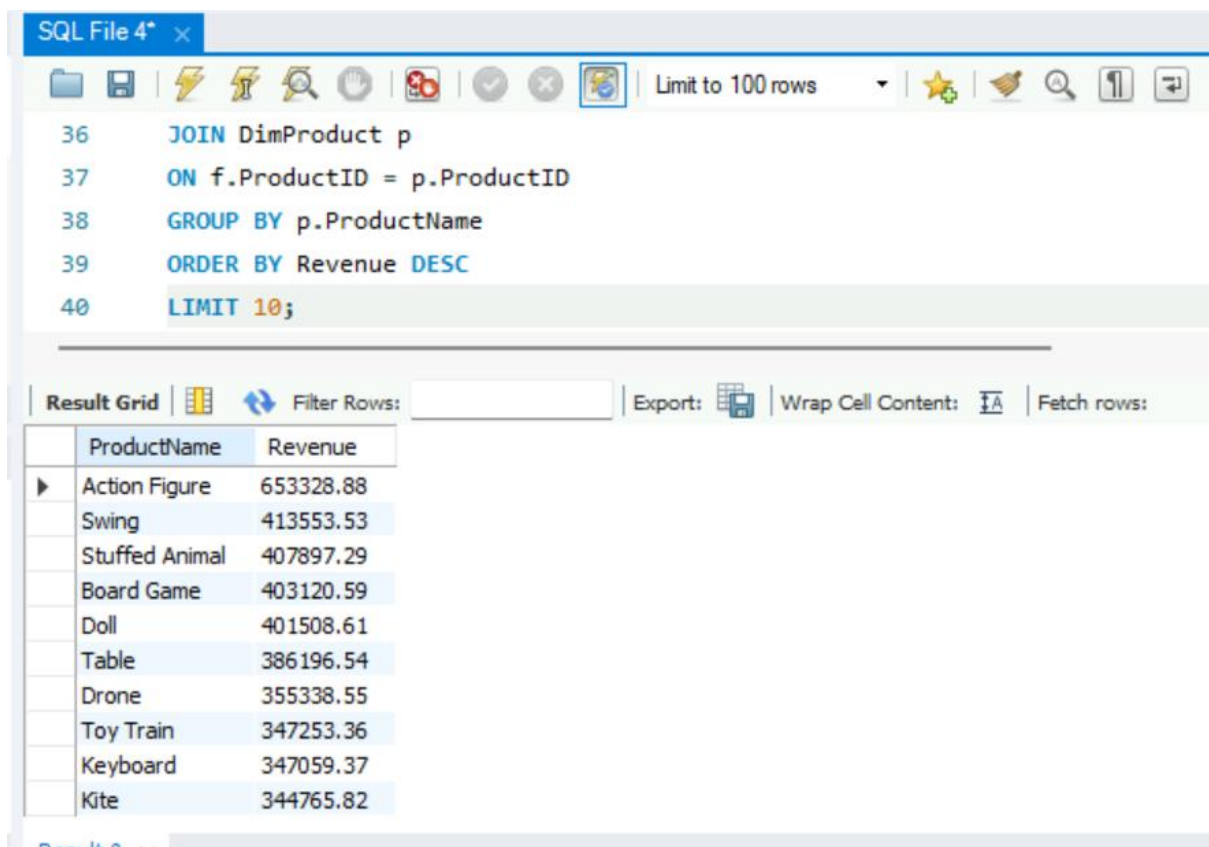
6.5 Sales by Product Category

```
26 FROM FactSales f
27 JOIN DimProduct p
28 ON f.ProductID = p.ProductID
29 GROUP BY p.Category
30 ORDER BY Revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Category	Revenue
▶	Toys	6107951.46
	Electronics	5237296.29
	Furniture	1538247.57
	Home Appliances	1276804.24

6.6 Top 10 Products



SQL File 4*

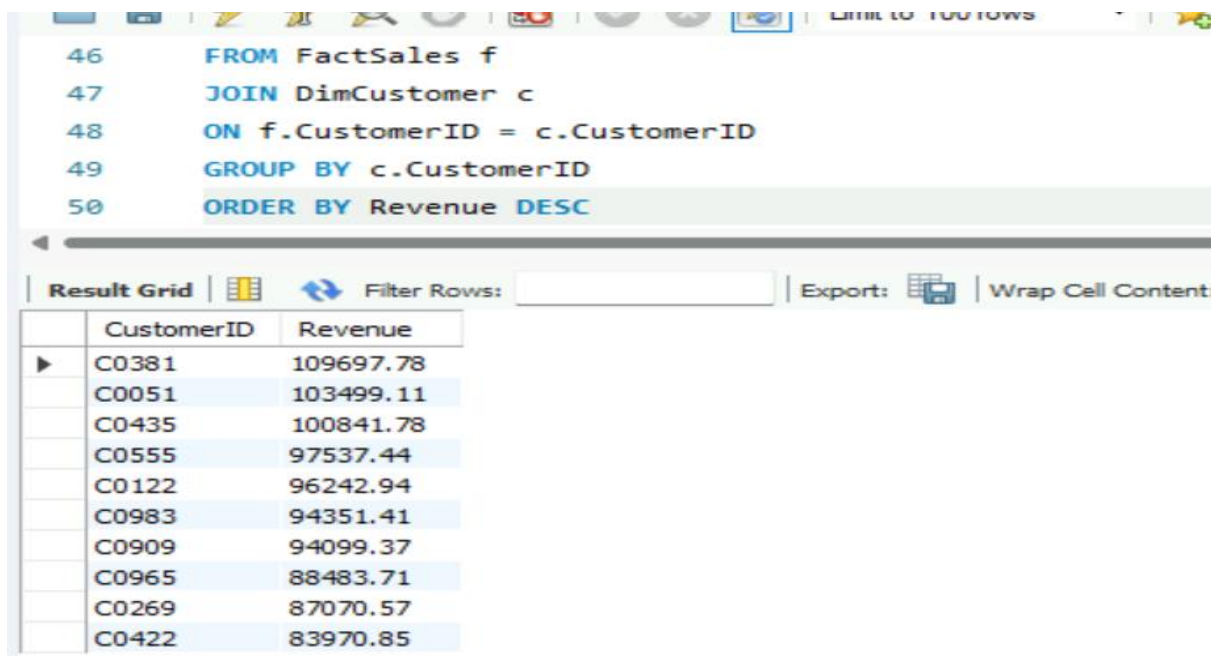
```
36 JOIN DimProduct p
37 ON f.ProductID = p.ProductID
38 GROUP BY p.ProductName
39 ORDER BY Revenue DESC
40 LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	ProductName	Revenue
▶	Action Figure	653328.88
	Swing	413553.53
	Stuffed Animal	407897.29
	Board Game	403120.59
	Doll	401508.61
	Table	386196.54
	Drone	355338.55
	Toy Train	347253.36
	Keyboard	347059.37
	Kite	344765.82

Result 2

6.7 Top Customers



SQL File 4*

```
46 FROM FactSales f
47 JOIN DimCustomer c
48 ON f.CustomerID = c.CustomerID
49 GROUP BY c.CustomerID
50 ORDER BY Revenue DESC
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	CustomerID	Revenue
▶	C0381	109697.78
	C0051	103499.11
	C0435	100841.78
	C0555	97537.44
	C0122	96242.94
	C0983	94351.41
	C0909	94099.37
	C0965	88483.71
	C0269	87070.57
	C0422	83970.85

6.8 Monthly Sales Trend

```
59 JOIN DimDate d
60 ON f.DateID = d.DateID
```

Result Grid   Filter Rows: Export:  Wrap Cell Content:			
	Year	MonthName	Revenue
▶	2024	June	2120907.19
	2024	July	2289080.72
	2024	August	2792319.57
	2024	September	2182497.38
	2024	October	2424428.35
	2024	November	2351066.35

6.9 Average Order Value

```
63
64 -- Average Order Value--
65 • SELECT
66     AVG(SalesAmount) AS AverageOrderValue
67 FROM FactSales;
```

Result Grid   Filter Rows: Export:  Wrap Cel	
	AverageOrderValue
▶	1416.029956

6.10 Chapter Conclusion

SQL analysis was performed on the retail data warehouse to gain insights into revenue, customer purchases, product performance, and sales trends.

Chapter 7: Power BI Dashboard Development

This chapter will include:

- Data Import from SQL
- Data Model Relationships
- DAX Measures
- KPI Cards
- Charts & Visuals
- Dashboard Design
- Business Insights

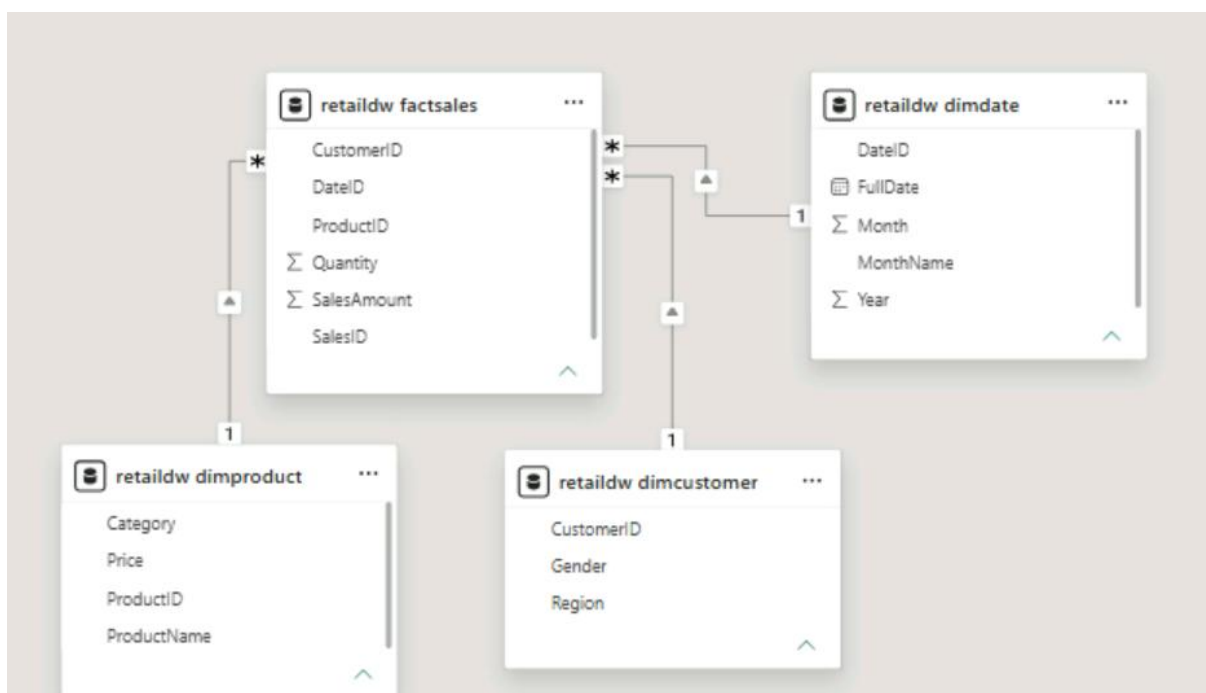
7.1 Objective

Power BI was used to visualize retail sales data and generate business insights through interactive dashboards.

7.2 Data Import

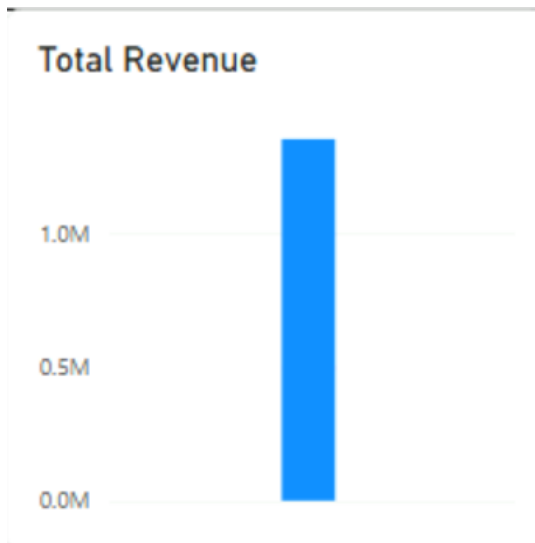
The warehouse tables were imported from SQL into Power BI for visualization and analysis.

7.3 Data Model

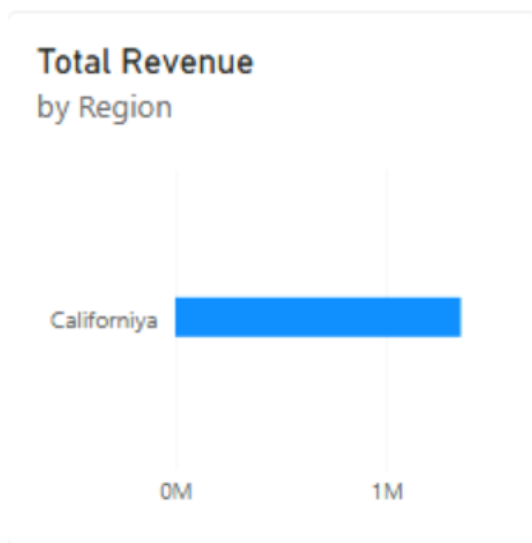


7.4 DAX Measures

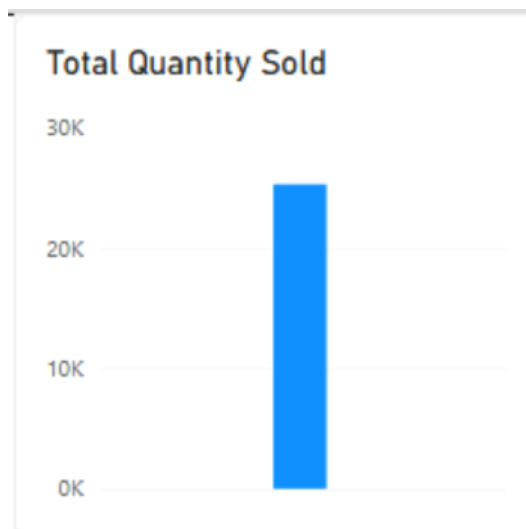
Total Revenue



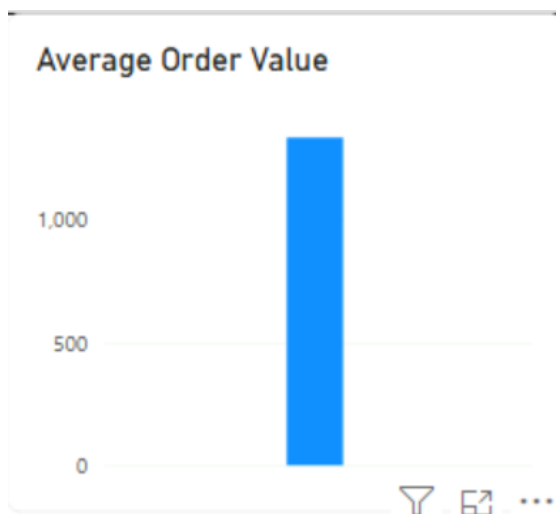
Total Orders



Total Quantity Sold



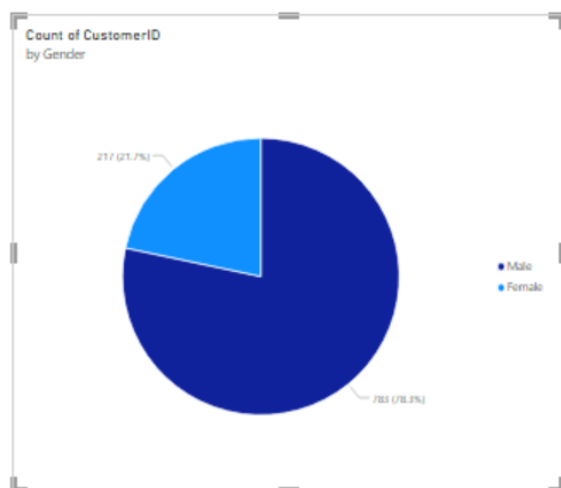
Average Order Value



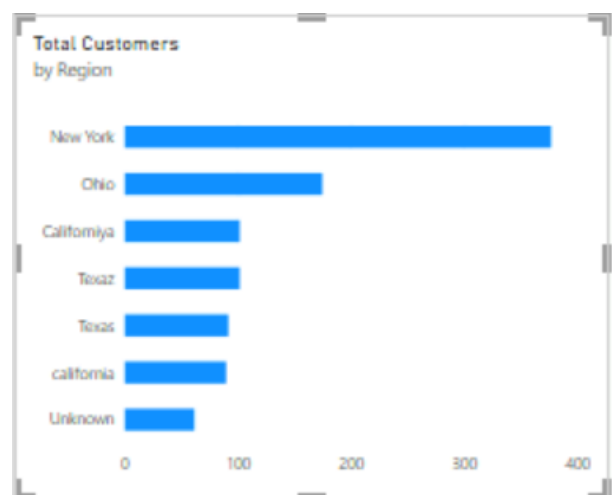
7.5 Dashboard Visualizations



Customer Demographics Analysis ||



Customers by Region



GitHub : [Case_Study/W1_Case_Study_2](https://github.com/amankumar1ext-web/Case_Study/tree/main/W1_Case_Study_2) at main · amankumar1ext-web/Case_Study

github.com/amankumar1ext-web/Case_Study/tree/main/W1_Case_Study_2

EOD